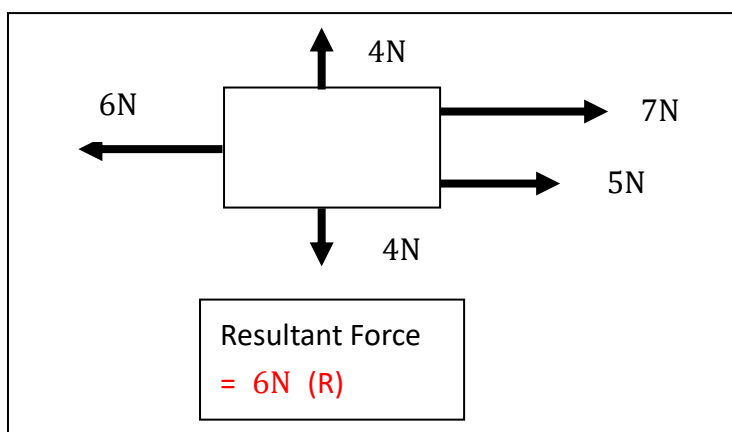
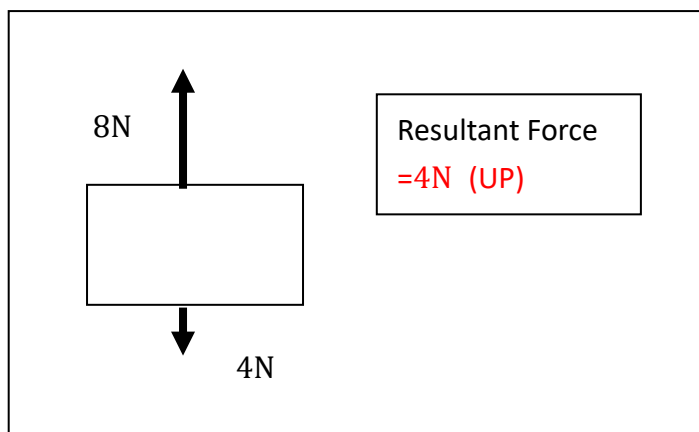
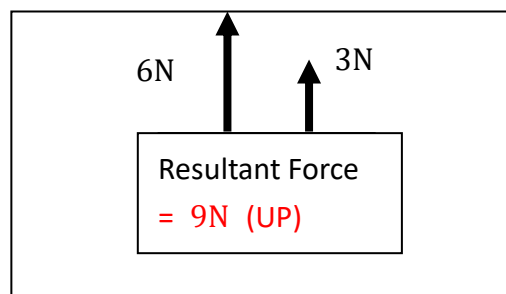
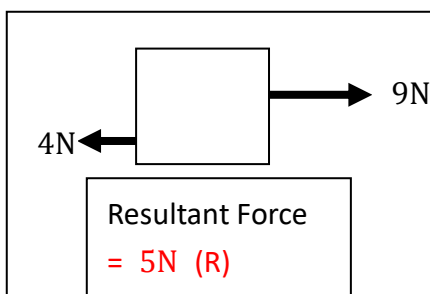
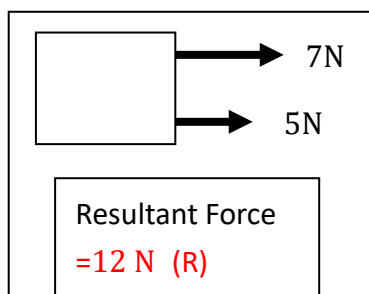
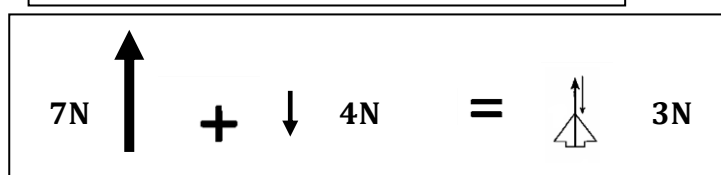
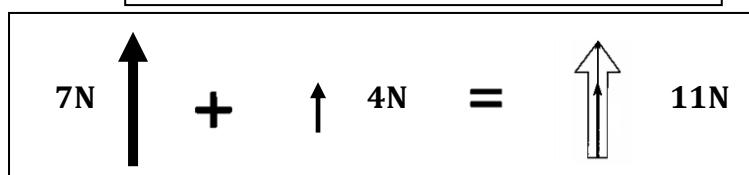
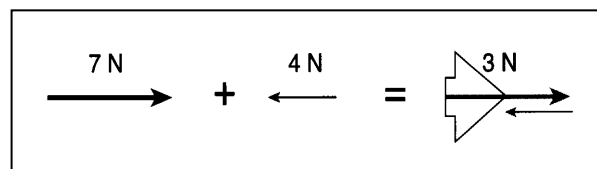
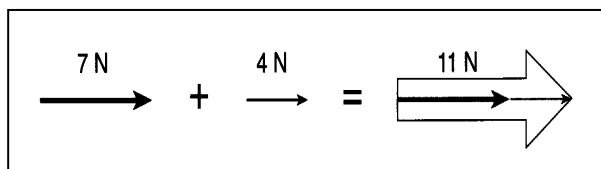


- **Force** is a **vector**. An arrow is used to represent a force in drawings to indicate its **direction** and **size**.
- A force can (i) speed up or slow down an object.
(ii) change the moving direction of an object.
(iii) change the shape of an object.
- Force can be measured by **spring balances** or **force sensors**.

Addition of forces along a straight line



NEWTON'S FIRST LAW

An object **remains at rest** or travel with **uniform velocity** unless act on by a net force (unbalanced force).

- When an object is at rest, it tends to remain at rest. When an object is moving, it tends to move in uniform velocity. This tendency is called **inertia**.
- The **greater** is the **mass** of a body, the **larger** is its **inertia**.

Remarks: No net force is needed to keep an object to move with uniform velocity.



1. The net force acting on an object is zero. The object may be

- (1) moving in a circle at a constant speed.
- (2) moving at a constant velocity.
- (3) at rest.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

C

2. Which of the following statements concerning the motion of an object is/are correct ?

- (1) A constant unbalanced force is needed to keep an object moving with uniform velocity.
- (2) An increasing unbalanced force is needed to keep an object moving with uniform acceleration.
- (3) An object may remain at rest if there is no unbalanced force acting on it.

- A. (2) only
- B. (3) only
- C. (1) & (2) only
- D. (1) & (3) only

✗	(1)	To keep an object moving with uniform velocity, i.e. zero acceleration, the net force should be zero. Thus no unbalanced force is needed.
✗	(2)	To keep an object moving with uniform acceleration, i.e. constant acceleration, the net force (unbalance force) should be constant, not increasing.
✓	(3)	If there is no unbalanced force, the acceleration is zero. The object may remain at rest OR may move with uniform velocity.

B

3. Which of the following statements for an object momentarily at rest is/are correct?

- (1) The speed of the object is zero
- (2) The net force acting on the object is zero
- (3) A force is required to change the speed of the object

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

- (1) ✓ Speed and velocity are zero at rest.
- (2) ✗ The object is just momentarily at rest. Its speed is changing. That means it has acceleration, which is brought about by a net force. An example is a ball thrown up to its highest position. Its speed is zero but there is a force (the weight of the ball) acting downward.
- (3) ✓ Without the action of a force, an object remains at rest or moves with constant speed.

C

4. In which of the following motions there must be an unbalanced force?

- (1) An object moves on a rough horizontal table with uniform velocity
- (2) An object remains stationary on a table moving with uniform velocity
- (3) An object falls freely in air

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

B
An unbalance force is required whenever there is acceleration in the motion. (1) and (2) are uniform motions but (3) has an acceleration due to gravity.

B

5. Which of the following statements involve Newton's first law of motion?

- (1) When a car stops suddenly, the passengers in the car are thrown forward
- (2) When the bowler releases a bowling ball from his hand, the ball will move with constant velocity on a smooth track
- (3) A spacecraft travels with uniform velocity in space even when its engine are shut off
- (4) A tennis ball changes its direction when it is hit by a tennis racket

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (4) only
- D. (1), (2) and (3) only

D

NEWTON'S SECOND LAW

The acceleration of a body is directly proportional to the unbalanced force acting on it and is inversely proportional to its mass.

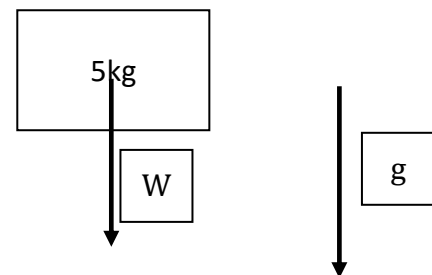
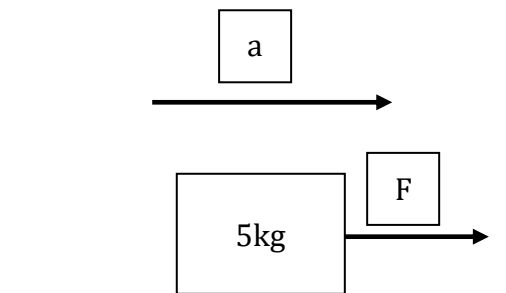
$$F_{\text{net}} = ma$$

Unit: Newton (N)

F = Force
m = mass
a = acceleration

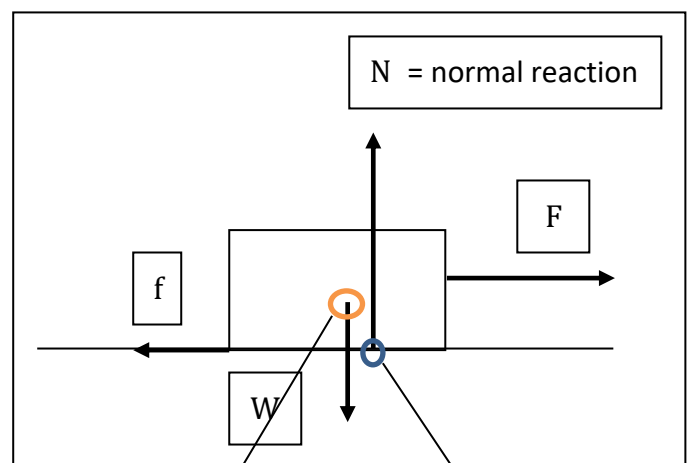
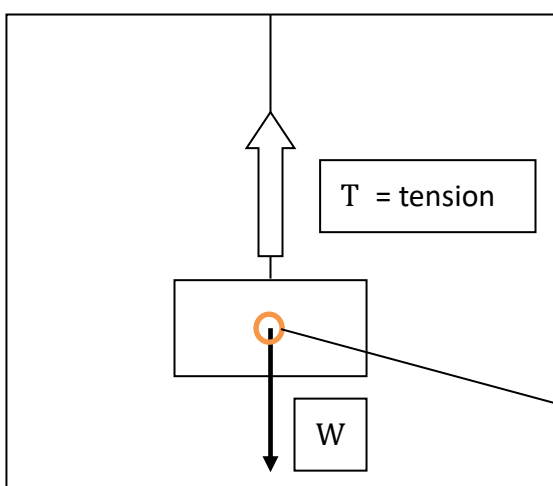
$$W = mg$$

W = weight
m = mass
g = gravitational acceleration



Free-body diagram (Force diagram)

of an object shows **ALL** the forces acting on the object.



Weight must be drawn from the **centre** of the object

Normal reaction must be drawn **from the contact surface** between the object and the supporting surface.

Remarks: Centre of gravity: The point that gravitational force(weight) act on the object

6. Which of the following pairs of quantities of an object ~~must be in the same direction?~~

- (1) displacement and velocity
(2) velocity and acceleration
(3) acceleration and resultant force
A. (2) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

- (1) * Consider a stone thrown upwards. When the stone is falling (but still above the original position), its displacement from the origin is pointing upwards, while its velocity downwards.
(2) * Consider the same example as in (1). When the stone is rising, its velocity is pointing upwards, while its acceleration downwards (acceleration due to gravity).
(3) ✓ By $F_{\text{net}} = ma$, the direction of the acceleration of an object is **always** the same as that of the resultant force acting on it.

B

7. Which of the following is a non-contact force (force acting on the object without contacting the object)?

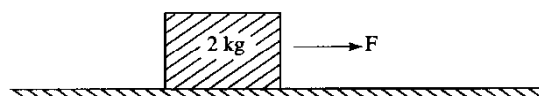
- A Tension in a rope
B Electric force between two electrons
C Friction between foot and ground
D Normal force acting on the desk by the ground

B

8. A block is pulled by a string on a smooth surface as shown below.

- (a) Find the acceleration of the block when F is 4 N.
(b) The string is broken when the block is moving. Find the acceleration of the block.

Answer : 2 ms^{-2}
Answer : 0



9. A toy car of mass 2 kg is at rest on a smooth horizontal table. If a force of 2 N acts on a toy car for 5 s, calculate its final velocity.

- A. 0 ms^{-1}
B. 0.5 ms^{-1}
C. 5 ms^{-1}
D. 50 ms^{-1}

$$\text{Acceleration } a = \frac{F}{m} = \frac{2}{2} = 1 \text{ m s}^{-2}$$

By $v = u + at$,
 $v = 0 + (1)(5)$
 $= 5 \text{ m s}^{-1}$

C

10. Mike pulls a block on the ground with a force of 50 N. The mass of the block is 8 kg and the friction acting on the block is 26 N. What is the acceleration of the block?

- A. 2 ms^{-2}
B. 3 ms^{-2}
C. 3.25 ms^{-2}
D. 6.25 ms^{-2}

By $F = ma$,

$$a = \frac{F}{m} = \frac{50 - 26}{8} = 3 \text{ m s}^{-2}$$



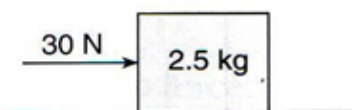
B

11. The acceleration of the 2.5 kg block is measured as 4 ms^{-2} . Find the friction between the block and the ground.

- A. 10 N
B. 12 N
C. 20 N
D. 30 N

$$30 - f = 4(2.5)$$

$$f = 20 \text{ N}$$



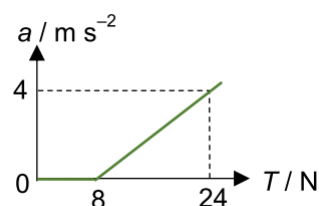
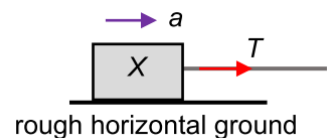
C

12. A block of mass M , initially at rest, is pulled by a horizontal string with tension T on a rough horizontal ground. Showing on the graph is the relation of tension T and acceleration a as T increases gradually from zero. Find M and the friction f on the block by the ground when T is 5 N.

A

- A $M = 4 \text{ kg}, f = 5 \text{ N}$
B $M = 4 \text{ kg}, f = 8 \text{ N}$
C $M = 6 \text{ kg}, f = 5 \text{ N}$
D $M = 6 \text{ kg}, f = 8 \text{ N}$

The block starts to move when tension $T > 8 \text{ N}$.
 $\therefore$ Maximum friction = 8 N
 $\therefore$ When T is 5 N, the block does not move and the friction f is 5 N.
When $T = 24 \text{ N}$, $a = 4 \text{ m s}^{-2}$.
By $F_{\text{net}} = ma$,
 $24 - 8 = M \times 4$
 $M = 4 \text{ kg}$



13. A force F is exerted on a block resting on a smooth surface as shown. After 2 s, another force having the equal magnitude but opposite direction is also applied. The block will

C

- A. come to rest immediately.
B. come to rest after 2 s.
C. move with a constant velocity.
D. move with an increasing velocity.



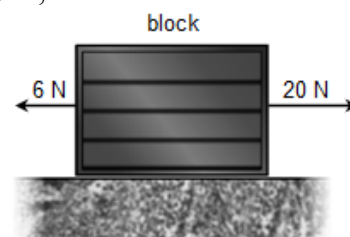
14. At the instant shown, the block is at rest on a rough surface, meanwhile it is acted on by two horizontal forces of magnitudes 6 N and 20 N.

Which of the following statements is/are correct?

- (1) The net force acting on the block is 14 N.
(2) When the force of magnitude 20 N is removed, the block moves to the right.
(3) When the magnitude of the force to the left increases from 6 N to 26 N, the block still remains at rest.

- A. (2) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

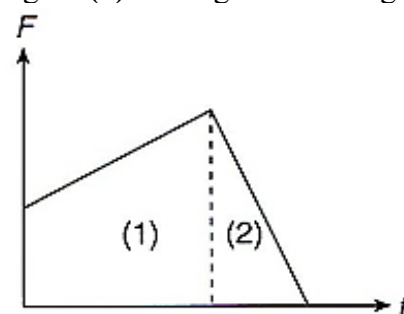
(1) It remain at rest Net force should be zero
(2) Friction = $20 - 6 = 14 \text{ N} > 6 \text{ N}$
(3) $26 - 20 = 6 \text{ N} < 14 \text{ N}$ It remain at rest



B

15. Determine the motion of an object, representing by region (1) and region (2) in the given $F - t$ graph.

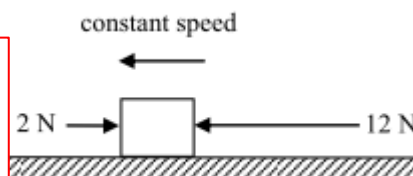
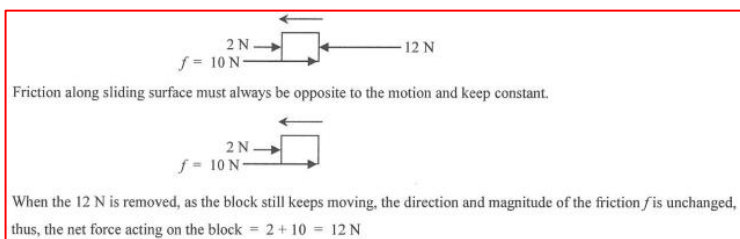
- | | Region (1) | Region (2) |
|----|---------------|---------------|
| A. | uniform speed | uniform speed |
| B. | decelerating | accelerating |
| C. | accelerating | decelerating |
| D. | accelerating | accelerating |



D

16. A block on a rough horizontal surface is moving to the left with constant speed under two horizontal forces 2 N and 12 N indicated as shown. If the force of 12 N is suddenly removed, what is the net force acting on the block at that instant?

- A. 12 N
B. 10 N
C. 8 N
D. 2 N



A

17. A block of mass 1 kg is moving to the left with an acceleration of 18 m s^{-2} when strings S_1 and S_2 are pulling it as shown. The tensions in S_1 and S_2 are 50 N and 20 N respectively.

S_1 is then suddenly broken. What will be the motion of the block immediately after this incident?

- A Stop immediately
B Slow down with 8 m s^{-2}
C Slow down with 32 m s^{-2}
D Speed up with 8 m s^{-2}

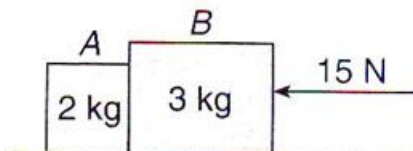
Take leftwards as positive.
Before S_1 is broken, by $F_{\text{net}} = ma$,
 $50 - 20 - f = 1 \times 18$
 $f = 12 \text{ N}$
Just after S_1 is broken, the block is moving to the left and so the friction points to the right.
By $F_{\text{net}} = ma$,
 $(-20) + (-12) = 1 \times a$
 $a = -32 \text{ m s}^{-2}$
 $\therefore$ The block will slow down with 32 m s^{-2} .



C

18. A force 15 N acts on two blocks as shown. Find the acceleration of A, and the force acting on A by B.

	a/ms^{-2}	f/N
A.	3	15
B.	7.5	15
C.	7.5	6
D.	3	6



D

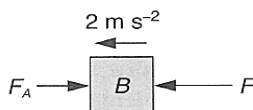
19. Two blocks A and B of masses 3 kg and 2 kg respectively are placed on a smooth horizontal table as shown in the figure. A horizontal constant force F is applied on A such that the two blocks move together towards the right with an acceleration of 2 ms^{-2} . If the force F is now applied on B such that the two blocks move together towards the left, find the force acting on B by A.

- A. 6 N towards the right
B. 6 N towards the left
C. 4 N towards the right
D. 4 N towards the left

The force F can be obtained as:

$$F = (3 + 2) \times 2 = 10 \text{ N}$$

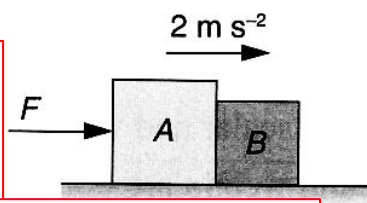
If the force F is applied on B, the magnitude of the acceleration is also 2 m s^{-2} .



Hence,

$$10 - F_A = 2 \times 2$$

$$\therefore F_A = 6 \text{ N towards the right}$$



A

20. A 1 kg block is placed on top of a 2 kg block on a smooth horizontal surface as shown. A horizontal force of 6 N is applied to the 2 kg block such that the blocks move together. What is the frictional force between the blocks?

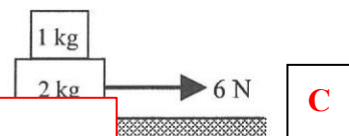
- A. 4 N
B. 3 N
C. 2 N
D. 0 N

Since the two blocks move together, they have the same acceleration.

Consider the two blocks as a whole system : $F = ma \quad \therefore (6) = (2 + 1) a \quad \therefore a = 2 \text{ m s}^{-2}$

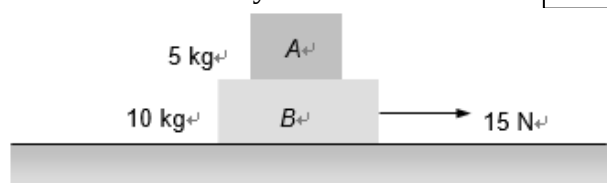
Consider the 1 kg block, the only force acting on it is the friction, thus friction provides the net force for acceleration.

$\therefore f = ma = (1)(2) = 2 \text{ N}$



21. Box A is placed on box B and the masses of A and B are 5 kg and 10 kg respectively. If all surfaces are smooth, what are the accelerations of A and B when 15 N applies on B horizontally?

	Acceleration of A	Acceleration of B
A	1 m s^{-2}	1 m s^{-2}
B	0	1.5 m s^{-2}
C	1.5 m s^{-2}	1.5 m s^{-2}
D	0	1 m s^{-2}



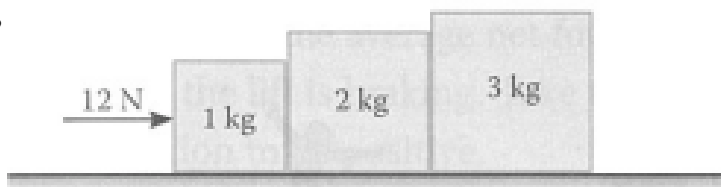
22. A horizontal force of 12 N pushes three boxes on a smooth horizontal ground as shown. The boxes have mass of 1 kg, 2 kg and 3 kg separately.

(a) Which box is exerted the greatest net force?

(b) Find the contact force between

(i) The 1 kg box and the 2 kg box

(ii) The 2 kg box and the 3 kg box



(a) The 3 kg box is exerted by the greatest net force.

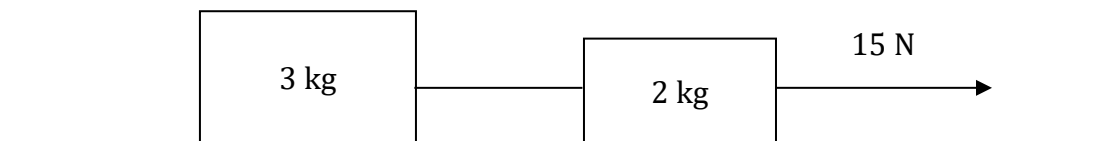
■ For the same a , $F \propto m$.

(b) (i) Applying $F = ma$, the common acceleration is $a = \frac{12}{1+2+3} = 2 \text{ m s}^{-2}$. (1M)

The contact force between the 1 kg and the 2 kg box is thus $(2+3) \times 2 = \boxed{10 \text{ N}}$. (1A)

(ii) The contact force between the 2 kg and the 3 kg box is thus $3 \times 2 = \boxed{6 \text{ N}}$. (1A)

23. Two blocks of mass 2 kg and 3 kg are tied together on a horizontal table. The system is pulled by a horizontal force of 15 N as shown below.



- (a) Find the acceleration of the system and the tension in the string connecting the blocks if the table is smooth.
(b) Find the acceleration of the system and the tension in the string connecting the blocks if the friction between the table and each block is 2 N.

Ans:

(a). $a = 3 \text{ ms}^{-2}$

$T = 9 \text{ N}$

Ans:

(b) By $F_{\text{net}} = ma$,

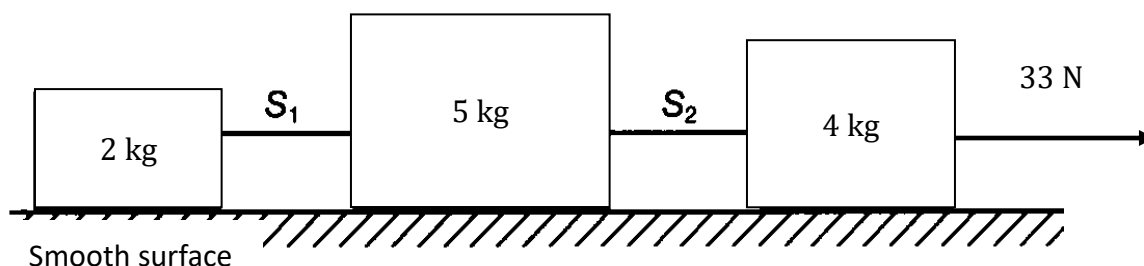
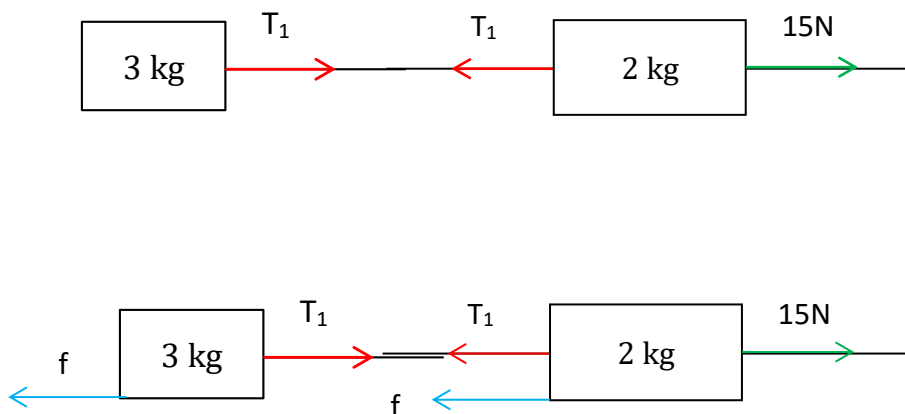
$15 - 2 - 2 = 5a$

$a = 2.2 \text{ ms}^{-2}$

$T_1 - f = ma$

$T_1 - 2 = 3(2.2)$

$T_1 = 8.6 \text{ N}$



24. (a) Find the acceleration of the 5 kg block.
(b) Find the tensions S_1 and S_2 .
(c) If the string S_1 suddenly breaks, describe the subsequent motions of the blocks. Support your answer with calculation.

ANS

Total mass = 11kg

$F_{\text{net}} = ma$,

$33 = 11(a)$, $a = 3 \text{ ms}^{-2}$

ANS

$T_{s1} = 2(3) = 6 \text{ N}$

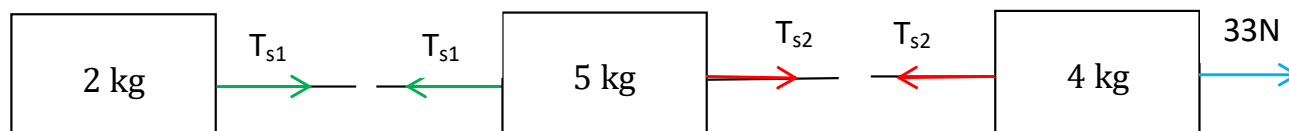
$T_{s2} - T_{s1} = 5(3)$,

$T_{s2} = 15 + 6 = 21 \text{ N}$

Ans

(c) The 2 kg block will move with constant velocity as net force acting on it became 0.

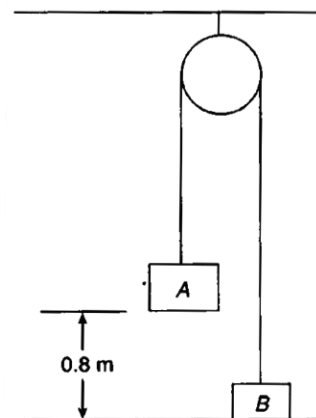
The 5 kg and 4 kg block will move with acceleration of 3.67 ms^{-2} .



警告：不可翻印，販賣，違者必究

25. Two blocks A and B of mass 2 kg and 1.5 kg respectively are connected by an inextensible string that passes through a smooth pulley. When B is stationary on the table and the string is taut, A is released from rest. Find the tension in the string and the acceleration of block A.

Ans
 $W_A - T = m_A a \rightarrow 20 - T = 2(a) \quad (1)$
 $T - W_B = m_B a \rightarrow T - 15 = 1.5(a) \quad (2)$
 $T = 17.14 \text{ N}, a = 1.43 \text{ m s}^{-2}$

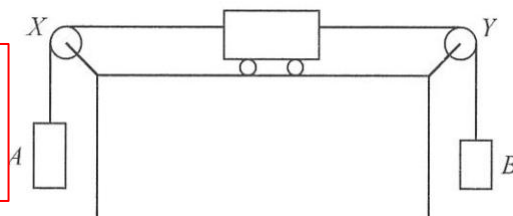


26. A trolley of mass 1 kg placed on a smooth horizontal table is connected by two light strings to blocks A and B of masses 0.75 kg and 0.25 kg respectively, as shown in the figure above. X and Y are frictionless pulleys. When the system is released, what will be its acceleration?

Consider the whole system of the two blocks and the trolley.

By Newton's second law: $F_{\text{net}} = m a$

$$(0.75 \times 9.81) - (0.25 \times 9.81) = (0.75 + 1 + 0.25) a \quad \therefore a = 2.45 \text{ m s}^{-2} \approx 2.5 \text{ m s}^{-2}$$



27. Blocks X and Y are connected by a light inextensible string passing over a fixed frictionless light pulley as shown. The mass of X and Y are 0.5 kg and 1 kg respectively. Initially, Y is 1 m above the ground and the string is taut. The system is then released from rest. What is the speed of Y just before it reaches the ground?

By conservation of energy,

$$m_Y g h = \frac{1}{2} (m_X + m_Y) v^2$$

$$(1)(9.81)(1) = \frac{1}{2} (0.5 + 1) v^2$$

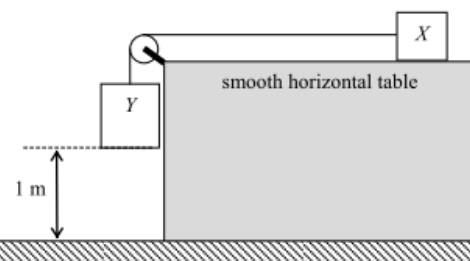
$$\therefore v = 3.62 \text{ m s}^{-1}$$

Alternatively, use Newton's 2nd law. Regard blocks X and Y as a single body performing an 1-D motion:

$$m_Y g = (m_X + m_Y) a \Rightarrow a = \frac{m_Y g}{m_X + m_Y}$$

Then, using $v^2 - u^2 = 2as$ and $u = 0$ gives $v = \sqrt{2as}$:

$$\therefore v = \sqrt{\frac{2m_Y g s}{m_X + m_Y}} = \sqrt{\frac{2(1)(9.81)(1)}{1.5}} = 3.62 \text{ m s}^{-1}$$



28. As shown below, three blocks X, Y and Z are connected with strings. The masses of X and Z are both 2 kg and the mass of Y is 1 kg. The frictional forces acting on X and Y are 5 N and 2.5 N respectively. Assume that the pulley is smooth.

(a) Find the magnitude of the acceleration of the blocks. (2 marks)

(b) Find the tension in the string connecting Y and Z. (2 marks)

(a) Consider the blocks as a whole.

By $F = ma$,

$$m_Z g - f_X - f_Y = (m_X + m_Y + m_Z) a$$

$$a = \frac{m_Z g - f_X - f_Y}{m_X + m_Y + m_Z} = \frac{2 \times 9.81 - 5 - 2.5}{2 + 1 + 2} = 2.42 \text{ m s}^{-2}$$

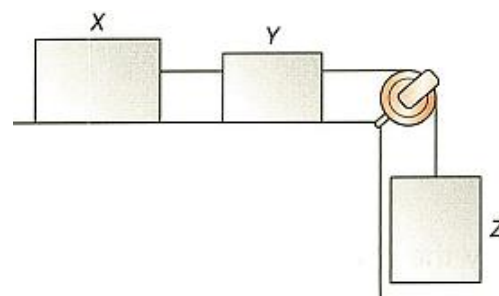
$\therefore$ The magnitude of acceleration is 2.42 m s^{-2} .

(b) Consider block Z.

$$m_Z g - T = m_Z a$$

$$T = m_Z g - m_Z a = 2 \times 9.81 - 2 \times 2.42 = 14.8 \text{ N}$$

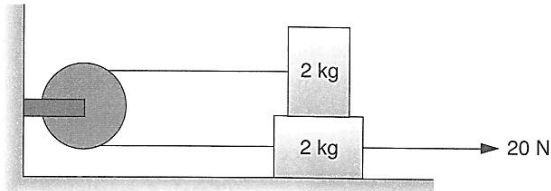
$\therefore$ The tension is 14.8 N.

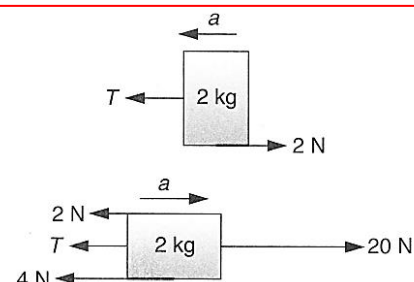


29. Two blocks of the same mass of 2 kg are placed on a rough horizontal surface. They are connected by an inextensible light string and the string passes over a smooth pulley fixing on a vertical wall as shown in the above figure. If a horizontal force of 20 N pulls the lower block and the friction between the two blocks is 2 N while the friction between the lower block and the horizontal table is 4 N.

(a) Draw force diagram for two blocks

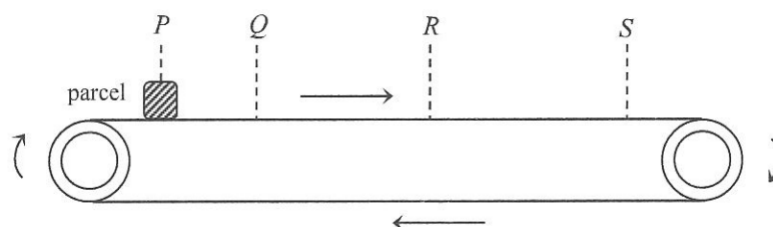
(b) Find the initial acceleration of the lower block.





$$\begin{cases} T - 2 = 2a & [1M] \\ 20 - 4 - 2 - T = 2a & [1M] \\ 12 = 4a & [1M] \\ \therefore a = 3 \text{ m s}^{-2} \text{ (to the right)} & [1A] + [1A] \end{cases}$$

30.



The above Figure shows a conveyor belt in a factory. A parcel of mass 10 kg is placed at position P when the belt remains at rest. The workman controls the belt such that the parcel undergoes a motion described in the below Table. The parcel and the conveyor belt move together without slipping during the entire motion.

Position of the parcel	Motion	Data given
$P \rightarrow Q$	Uniform acceleration	$PQ = 5 \text{ m}$ and time required = 2 s
$Q \rightarrow R$	Uniform velocity	-----
$R \rightarrow S$	Uniform deceleration to rest	-----

(a) Consider the motion when the parcel travels from P to Q.

(i) Draw a free-body diagram to show all forces acting on the parcel in the space provided below. Name the forces.

(2 marks)

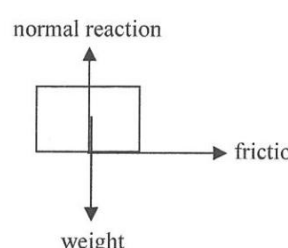
(ii) Find the net force acting on the parcel.

(3 marks)

(b) In the Figure below, sketch a graph to show the variation of the frictional force exerted by the conveyor belt on the parcel.

(3 marks)

(a) (i)



(ii) By $s = ut + \frac{1}{2}at^2$

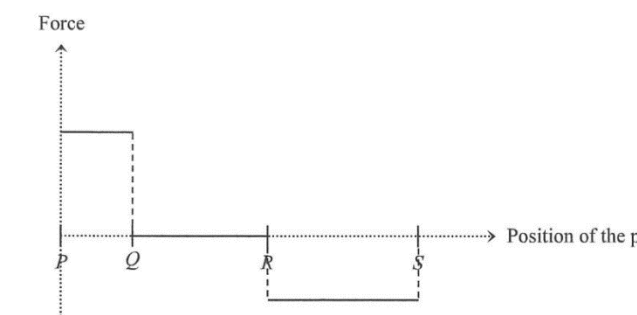
$$\therefore (5) = (0) + \frac{1}{2}a(2)^2$$

$$\therefore a = 2.5 \text{ m s}^{-2}$$

$$F = ma$$

$$= 10 \times 2.5 = 25 \text{ N}$$

(b)

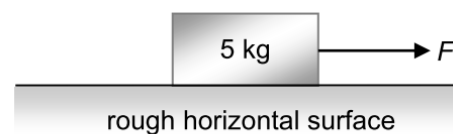


< P to Q : a horizontal line above or below the x-axis >

< Q to R : a horizontal line at the x-axis >

< R to S : a horizontal line at the opposite side of the line PQ >

31. (a) A block of mass 5 kg is placed on a **rough** horizontal surface (Fig a). A horizontal force F of 5 N is applied on it. The block moves at a constant velocity of 1 m s^{-1} to the right.
- (i) Find the net force acting on each block. (3 marks)
- (ii) Find the force acting on the 5-kg block by the 10-kg block. (1 mark)



(a) (i) 5 N	1A
(ii) Take rightward as positive. By $F_{\text{net}} = ma$, $10 - 5 = 5 \times a$ $a = 1 \text{ m s}^{-2}$ The acceleration of the block is 1 m s^{-2} (rightwards).	1M 1A

- (b) The 5-kg block and another 10-kg block are then placed together on a **smooth** horizontal surface (Fig b). A 75-N force is applied horizontally towards the right against the 5-kg block.
- (i) Find the net force acting on each block. (3 marks)
- (ii) Find the force acting on the 5-kg block by the 10-kg block. (1 mark)

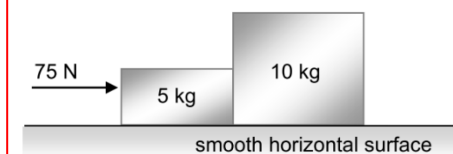
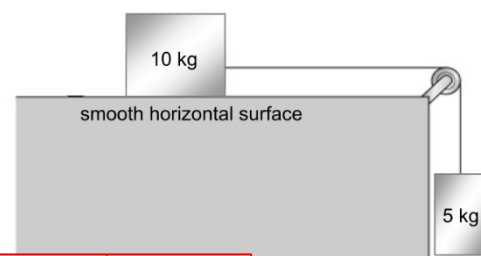


Fig b

(b) (i) Take rightward as positive. Consider the system of the two blocks. By $F_{\text{net}} = ma$, $75 = (5 + 10) \times a$ $a = 5 \text{ m s}^{-2}$ Consider 5-kg block. $F_{\text{net}} = ma = 5 \times 5 = 25 \text{ N}$ Consider 10-kg block. $F_{\text{net}} = ma = 10 \times 5 = 50 \text{ N}$	1M 1A 1A 1A
(ii) 50 N	1A

- (c) The blocks are then rearranged as shown in Figure c.
- (i) Find the magnitude of the acceleration of the blocks. (2 marks)
- (ii) A student says that the acceleration of the 10-kg block will be the same as in (i) if the 5-kg block is replaced by a pulling force of $5g$ vertically downwards, where g is the acceleration due to gravity. Comment on this statement. (3 marks)



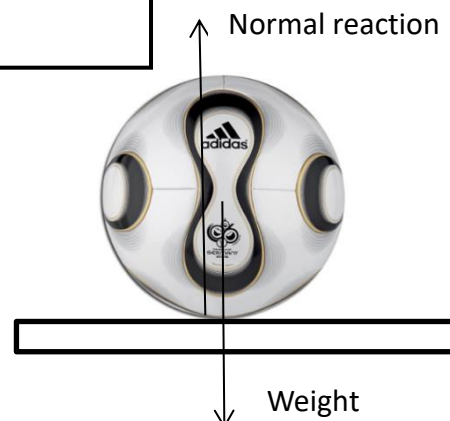
(c) (i) Consider the system of the two blocks. By $F_{\text{net}} = ma$, $5 \times 9.81 = (10 + 5) \times a$ $a = 3.27 \text{ m s}^{-2}$	1M 1A
(ii) The statement is not correct. In (i), the weight of the 5-kg block is used to accelerate both blocks. On the contrary, the $5g$ force is used to accelerate the 10-kg block only, so the acceleration of the 10-kg block will be greater.	1A 1A 1A

Newton's third law of motion

To every action, there is always an equal and opposite reaction exists. The action and reaction forces act on different interacting objects simultaneously

Identifying action and reaction pairs:

- Same magnitude
- Opposite in direction
- Act on two different objects



32. 2005/II/29

Which of the following pairs of forces is an action and reaction pair?

- A. The force exerted by the string on X **and** The force exerted by the string on Y
- B. The force exerted by the string on Y **and** The force exerted by Y on the string
- C. The gravitational force exerted on Y **and** The force exerted by X on the horizontal surface
- D. The gravitational force exerted on Y **and** The force exerted by the string on Y

B

33. 2003/II/12

A child is sitting on a chair as shown above. Which of the following pairs of forces is/are (an) action and reaction pair(s)?

- | | | |
|---|------------|---|
| (1) The gravitational force exerted by the earth on the child | and | The normal reaction exerted by the chair on the child |
| (2) The force exerted by the child on the chair | and | The normal reaction exerted by the chair on the child |
| (3) The force exerted by the chair on the ground | and | The gravitational force exerted by the earth on the chair |



B

- A. (1) B. (2) C. (1) and (3) D. (2) and (3)

34. A block is pulled by a string on a table. The above figure shows the tension and friction on the block. Which of the following is the reaction to the friction on the block?

- A. The tension from the string on the block
- B. The friction from the block on the table
- C. The normal reaction from the table on the block
- D. There is no reaction to the friction



B

35. A man of weight W is standing on the ground. Which of the followings forms an action-and-reaction pair with W ?

- A. The reaction force acting on the man by the ground.
- B. The force acting on the ground by the man.
- C. The gravitational force acting on the man by the Earth.
- D. The gravitational force acting on the Earth by the man.

N = 由地面施於人的力
(Force acting on the man by the ground)
 $= \text{ground} F_{\text{man}}$

R = 由人施於地面的力
(Force acting on the ground by the man)
 $= \text{man} F_{\text{ground}}$

W = 人的重量 (weight)
 $= \text{earth} F_{\text{man}}$

D